An INR-Native Climate-Fintech Platform for Carbon Credit Trading, ESG Compliance and Fair, Trust-Based Farmer Allocation

PROJECT SYNOPSIS

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BACHELOR OF TECHNOLOGY

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Information Technology

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Supervisor's Verification

This is to verify that the synopsis entitled **"An INR-Native Climate-Fintech Platform for Carbon Credit Trading, ESG Compliance and Fair, Trust-Based Farmer Allocation"** is a bonafide part of the final year project work being carried out by Danish Shanil Shah student of B.Tech in **Information Technology**, under my supervision.

The content presented in the synopsis is in accordance with the academic standards and guidelines of the department.

Full Signature of the Candidate (with date)

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1 Abstract

Empirical research shows that financial transparency is a major weakness in carbon-credit projects, while farmers trust State Agriculture Departments and local extension networks more than private or international organizations. Our system is an INR-native climate-fintech platform designed around these findings. It uses trusted local institutions for field verification, auditable DBT-style settlement, and a Farmer Reliability Point System (FRPS) based on eight trust-related constructs. A Fair Credit Allocation Engine (FCAE) then distributes bulk corporate ESG demand across eligible farmers through an efficient, starvation-free algorithm. By integrating trust and transparency directly into the system, our system aims to provide a practical and farmer-centric carbon-credit platform.

2 Literature Review

Recent research on agricultural carbon markets establishes that successful implementation requires a foundation of farmer participation, effective carbon-farming practices, reliable measurement and verification (MRV), and transparent market infrastructure. Central to this foundation is farmer participation, which Block et al. [2] and Cammarata et al. [3] demonstrate is heavily influenced by payout conditions, perceived repayment risks, knowledge, and institutional trust. Building on this, Mallappa et al. [1] found that Indian farmers exhibit higher trust in State Departments and local extension networks compared to private entities, while identifying financial transparency as a critical weakness in existing systems. To generate viable credits within such trusted frameworks, literature confirms that conservation agriculture drives essential soil-carbon sequestration [5, 7, 8, 14], provided there are appropriate incentives, additionality, and equitable participation [11, 12, 15]. However, quantifying these environmental benefits introduces the challenge of MRV; Okoli and Birkenberg [9] outline the trade-offs between accuracy, cost, and scalability, whereas Li et al. [10] and Ghimirey et al. [16] highlight the efficacy of satellite remote sensing and machine learning in mitigating site-specific measurement uncertainties. Currently, studies generally treat these dimensions—farmer trust, MRV, and digital market mechanics—as isolated issues rather than an integrated ecosystem. Translating these fragmented requirements into a cohesive solution, our system bridges the research gap through an INR-native architecture that combines satellite-assisted MRV, trusted institutional verification, and auditable settlement, governed by measurable software mechanisms like the Farmer Reliability Point System (FRPS) and the Fair Credit Allocation Engine (FCAE).

3 Problem Definition

Empirical evidence from farmer-trust research shows that India's rural climate finance ecosystem suffers from four structural failures:

1. **Financial Opacity:** financial transparency is consistently the weakest-rated dimension of trust

across all carbon-project stakeholders, and a large share of surveyed carbon farmers report receiving no monetary benefit at all, directly driving disadoption.

2. **Misplaced Trust Channels:** many carbon-credit projects route farmer engagement through NGOs, private companies, or international organisations – exactly the actors farmers trust *least* – while under-utilising State Departments of Agriculture, KVKs, and progressive farmer networks, which farmers trust *most*.

3. **INR Exclusion from Carbon Markets:** existing carbon platforms price and settle in USD, exposing farmers to foreign-exchange volatility with no rupee-native, DBT-style settlement pathway.

4. **No Fair Allocation Mechanism:** when a corporate buyer places a bulk credit order, no existing system guarantees a fair, starvation-free distribution of that demand across many eligible farmers.

The specific problem this project targets is the design and implementation of a climate-fintech platform that (a) routes verification, training, and payment communication through the locally trusted channels identified above, (b) makes financial transparency an engineered, auditable system property rather than a policy promise, and (c) provides a computationally efficient, reputation-gated fair-allocation mechanism (FCAE, gated by FRPS) for distributing bulk corporate demand – making the platform both more trustworthy to farmers and more operationally feasible to deploy.

4 Methodology / Planning of Work

The proposed system follows a four-stage pipeline, illustrated in Fig. 1.

1. **Farmer Onboarding and Land Data:** Farmers complete KYC and provide bank/UPI details. Land parcels and crop information are geo-tagged and securely recorded.

2. **MRV and Carbon Calculation:** Sentinel-2 [3], Landsat-9 and Bhuvan data are combined with IPCC emission factors [10] and FAO crop references [7] to estimate sequestered CO_2e. Results are field-verified through State Agriculture Departments and local KVKs.

3. **Credit Issuance and Registry:** Verified credits are recorded on Polygon [12] as a minimal proof containing credit ID, CO_2e, vintage year and farmer wallet. The process follows digital MRV principles [6] and VM0042 guidelines [17]. The FRPS is updated using trust-related factors identified by Mallappa et al. (2025).

4. **Corporate ESG Purchase and Allocation:** Corporate demand follows BRSR [16], CCTS [1,2,5] and GHG Protocol principles [9]. FCAE distributes demand among eligible farmers using a capped, FRPS-tiered round-robin algorithm with $O(N \log N)$ complexity. Distributed locking [15] prevents race conditions, followed by auditable INR settlement and farmer-visible payment records.

The FRPS/FCAE mechanism will be evaluated against first-come-first-served and reputation-based allocation using Gini-coefficient inequality, starvation rate, allocation latency, and order-fulfilment completeness.

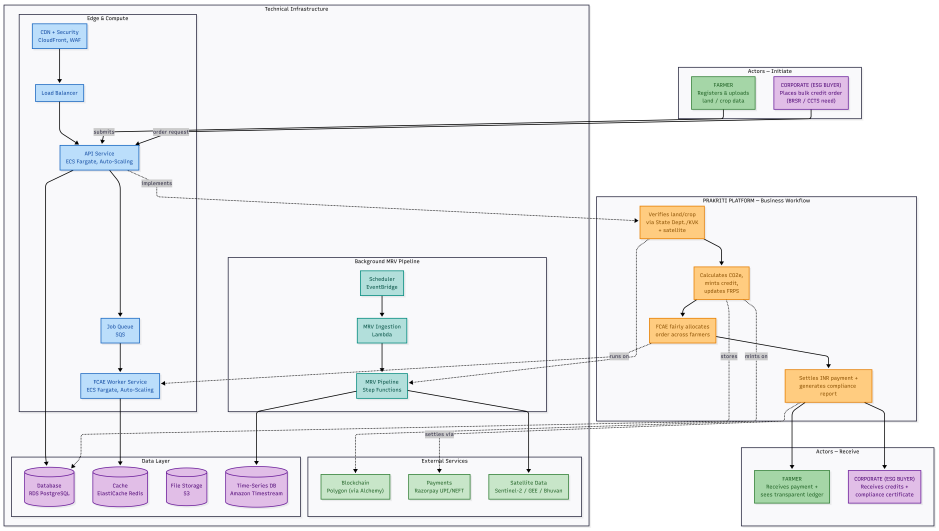


Figure 1: Proposed workflow of the our platform, from farmer onboarding to corporate ESG settlement.

5 Facilities Required for Proposed Work

5.1 Software

- Frontend: Next.js 14, TypeScript, Tailwind CSS (PWA)
- Backend: Node.js / Next.js API routes
- Database: PostgreSQL 16.x; TimescaleDB (satellite time-series data)
- Cache / Queues: Redis 7.x (FCAE locks and job queues)
- Offline Sync: Firebase Firestore, Firebase Storage
- **Satellite / MRV:** Google Earth Engine, Sentinel-2 [3], Landsat-9 and Bhuvan (ISRO); crop-level emission benchmarking using Indian studies [11,14], grid emission factors [4], and national agricultural GHG estimates [8].
- Payments: Razorpay (UPI / NEFT)
- Smart Contracts: Solidity, Hardhat, Polygon Amoy testnet

5.2 Hardware

- Development workstation(s) with minimum 8 GB RAM
- Cloud compute/hosting instance for backend and database services
- Internet connectivity for satellite data retrieval and blockchain interaction

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